

PAYEL BHATTACHARJEE

Tucson, AZ, USA

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EDUCATION

University of Arizona , Tucson, AZ, USA	Aug 2022 – Present
Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering	
School of Electrical, Computing, and Software Engineering	GPA: 3.79/4.00
Advisor: Dr. Ravi Tandon	
Kalinga Institute of Industrial Technology (KIIT) , Bhubaneswar, India	2017 – 2021
B.Tech. in Electronics & Telecommunication Engineering	GPA: 9.43/10.0

RESEARCH INTERESTS

Trustworthy Machine Learning, Large Language Model Alignment and Evaluation, Reward Modeling and Preference Learning, Differential Privacy-Preserving Machine Learning, Efficient LLM Inference, Conformal Prediction and Uncertainty Quantification, Causal Discovery and Causal Machine Learning.

PUBLICATIONS

Large Language Models, Alignment, and Efficient Inference

1. **P. Bhattacharjee**, R. Tandon. *Adaptive Multi-Value Control in LLMs via Causal Activation Steering*. arXiv preprint, 2026. **Under Review**.
2. **P. Bhattacharjee**, O. Simeone, R. Tandon. *Conformal Speculative Decoding*. Accepted for presentation in *ASILOMAR 2026* Conference (full paper forthcoming).
3. **P. Bhattacharjee**, O. Simeone, R. Tandon. *MARS: Margin and Semantic-Aware Data Augmentation for Reward Modeling*. arXiv preprint, 2026. **Under Review**.
4. N. Teku, F. Tian, **P. Bhattacharjee**, S. Chakraborty, A. Bedi, R. Tandon. *PROPS: Progressively Private Self-alignment of LLMs*. *Transactions on Machine Learning Research (TMLR)*, 2025.
5. **P. Bhattacharjee***, F. Tian*, M. Zhong*, G. Zhang, O. Simeone, R. Tandon. *Conformal Sparsification for Bandwidth-Efficient Edge-Cloud Speculative Decoding*. *NeurIPS AI4NextG Workshop*, 2025.

Privacy-Preserving Machine Learning and Causal Inference

6. **P. Bhattacharjee**, F. Tian, G. Rubin, J. Lo, N. Merchant, H. Hanson, J. Gounley, R. Tandon. *Learning to Diagnose Privately: DP-Powered LLMs for Radiology Report Classification*. *IEEE Access*, 2026.
7. F. Tian, **P. Bhattacharjee**, H. Hanson, G. Rubin, J. Lo, R. Tandon. *STAMP: Selective Task-Aware Mechanism for Text Privacy*. *European Chapter of the Association for Computational Linguistics (EACL)*, 2026.
8. **P. Bhattacharjee**, R. Tandon. *CURATE: Scaling Up Differentially Private Causal Graph Discovery*. *Entropy*, 2024.
9. **P. Bhattacharjee**, R. Tandon. *Adaptive Privacy for Differentially Private Causal Graph Discovery*. *IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2024.
10. Z. Guan, Z. Zhao, F. Tian, D. Nguyen, **P. Bhattacharjee**, R. Tandon, B. A. Prakash, A. Vullikanti. *A Framework for Multi-source Privacy Preserving Epidemic Analysis*. arXiv preprint, 2025.

Other Publications

11. **P. Bhattacharjee**, R. Garain, A. Basak, U. Singh. *Numerical Modelling and Performance Evaluation of SnS-Based Heterojunction Solar Cell with p^+ -SnS BSF Layer*. *Optical and Quantum Electronics*, 2022.
12. **P. Bhattacharjee**, S. Kar, N. Rout. *Sleep and Sedentary Behavior Analysis from Physiological Signals Using Machine Learning*. *IEEE 2nd International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)*, 2020.

*Equal contribution.

RESEARCH EXPERIENCE

University of Arizona

Tucson, AZ

Graduate Research Assistant/Associate — Advisor: Dr. Ravi Tandon

Aug 2022 – Present

- Developed **adaptive multi-value activation steering methods for large language models** that monitor value-specific representations during generation and dynamically coordinate multiple steering objectives.
- Proposed **margin- and semantic-aware reward modeling** methods that identify difficult preference comparisons and selectively augment them, improving reward learning under limited and noisy human-feedback data.
- Developed **differentially private LLM alignment methods** for learning from sensitive preference labels while providing formal privacy guarantees and preserving alignment performance.
- Developed **conformal speculative decoding frameworks for efficient LLM inference**, including methods for predicting context-dependent acceptance lengths and reducing edge-cloud communication with statistical reliability guarantees.
- Designed **adaptive differential privacy mechanisms for causal graph discovery** that allocate privacy budgets according to the informativeness and difficulty of causal tests, improving graph recovery under fixed privacy constraints.
- Developed **privacy-preserving learning methods for sensitive healthcare data**, including differentially private LLM-based radiology classification and multimodal diagnostic learning.

INDUSTRY EXPERIENCE

Bosch Global Software Technologies (BSW)

India

Associate Software Engineer

Jul 2021 – Jun 2022

- Developed Python-based software and automation tools for Advanced Driver Assistance Systems (ADAS) data-analysis workflows and data-processing and machine-learning solutions for automotive projects involving clients in the United States and Europe.

Bosch Global Software Technologies (BSW)

India

Project Intern

Mar 2021 – Jun 2021

- Developed Python and machine-learning tools for automated data analysis and engineering workflows.

TEACHING EXPERIENCE

University of Arizona

Tucson, AZ

Graduate Teaching Assistant, ECE 372

Aug 2022 – May 2023

I served as a Teaching Assistant for the ECE372 course, managing a class of 60 students. In this role, I oversaw laboratory sessions, guided students through experiments, and provided hands-on support in debugging and refining their implemented code. Additionally, I coordinated classroom activities, ensured smooth execution of lab exercises, and facilitated effective learning by addressing individual and group challenges. This experience strengthened my skills in both technical mentorship and classroom management.

AWARDS & HONORS

GPSC Travel Grant, University of Arizona

2024, 2025

People's Choice Award, ECE Graduate Poster Symposium, University of Arizona

2024

KIIT Merit Scholarship, KIIT University

2018

ACADEMIC SERVICE

Officer: Electrical and Computer Engineering Graduate Student Association, University of Arizona

2024 – 2026

Reviewer: IEEE ITW 2026, IEEE Transactions on Information Theory, EACL 2026

TECHNICAL SKILLS

Machine Learning: Large Language Models, Transformers, RLHF, Reward Modeling, Preference Learning, Differential Privacy, Causal Inference, Causal Discovery, Conformal Prediction, Federated Learning, Speculative Decoding.

Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA.

Programming: Python, C/C++, R, MATLAB.

Computing: GPU Computing, HPC Systems, Multi-GPU Training.